

WHY **STB** MANUFACTURING IS NOT PICKING UP IN INDIA

With the current trend indicating a drop in CRT TV and rise in LCD TV sales, why aren't Indian manufacturers moving in fast to churn out set-top boxes? Here's a look at why imports still rule the roost



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Set-top box (STB) is a device which decodes digital pictures and converts them into an analogue format, streaming the signal to the analogue TV. STBs can also transmit digital pictures using a high-definition multimedia interface (HDMI) connector.

Different types of STBs include direct-to-home (DTH) satellite boxes, digital cable boxes, digital terrestrial boxes and Internet protocol TV (IPTV). Different flavours of these STB categories include:

1. Simple zapper boxes
2. Personal video recorder (PVR) boxes with time-shift recording features, or live TV features like the PVRs from Tata Sky or Hathway cable
3. Hybrid boxes with IPTV
4. Boxes with return path feature

through various technologies like Ethernet, Docsys, GSM/GPRS or analogue modem technology

STB technology

In the past three years, there has been a phenomenal shift from analogue cable signals to digital transmission using STB as the underlying technology. Growth in the developed world, and the US in particular, has been tremendous with around 20 million STBs added per year. Focus has now shifted to Asian economies like India and China as these countries show enormous growth potential. China is expected to add around 17 million STBs in 2009, while India is expected to add 10 million. Currently there are about 700,000 STBs running on a subscription basis every month in India.

STB markets in Europe, Japan, the US and other developed nations are

moving towards high-definition TV (HDTV) and PVRs. With HDTV, consumers enjoy excellent picture quality. Once addicted such consumers can never settle for standard definition (SD) quality. In the US around 150 HD channels are beamed by direct TV and dish network and more are getting added.

PVR is another technology wherein customers can record a programme while they watch another programme on a different channel. PVR is becoming very popular in Europe and the US where its growth has surpassed even the high growth recorded by the zapper boxes market.

Beaming HD channels need substantial bandwidth on the satellite transponders—it takes a capacity of four SD channels to beam one HD channel.

Technology advancement to match demand

Fig. 1 indicates the ever-increasing demand for STBs. This is not possible without technical improvements in the underlying semiconductor technologies and software to enable easy integration of the conditional access system (CAS) and middleware.

Today we see a shift from DVB-S to DVB-S2 or DVB-T to DVB-T2. Another move is from MPEG-2 to MPEG-4 encoding (a higher compression scheme). This helps operators to add more channels to the existing transponder bandwidth. For example, based on DVB-S modulation and MPEG-2 encoding,